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Basics of Printing Processes

Course Code: 2101 | Semester: 2 | Credits: 3 | Rev: 2021

Course Overview

Teaching Scheme

- Lecture: 3 Hours/Week
- Tutorial: 0 Hours/Week
- Practical: 0 Hours/Week
- Total Periods: 45

Course Objectives

- Understand stages of printing process.
- Learn working principles of conventional methods.
- Identify basic units of printing machines.
- Explore digital printing technologies.

Course Outcomes (CO)

CO No.	Description	Cognitive Level
CO1	Explain the stages of printing process.	Understanding
CO2	Interpret basic types of printing processes.	Understanding
CO3	Describe the workflow of Letterpress, Offset, Flexography and Gravure.	Understanding
CO4	Describe features of screen-printing and Digital printing process.	Understanding

Module 1: Stages and History of Printing

Duration: 10 Hours | CO1

1.1 Definition of Printing

Printing is the process of reproducing text and images, typically with ink on paper using a print press. It is a mass-production industrial process and is an essential part of publishing and transaction printing.

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1.2 History of Printing

The journey of printing evolved from manual techniques to high-speed digital systems:

- **Early Stone Prints:** Ancient civilizations used seals and stones to stamp images.
- **Woodblock Printing:** Originating in China, images were carved into wooden blocks.
- **Movable Type:** Johannes Gutenberg (1440) revolutionized printing with metal movable type.
- **Industrial Revolution:** Steam-powered presses increased speed significantly.
- **Modern Era:** Introduction of Offset, Gravure, and Digital printing.

1.3 Classification of Print Works

The printing workflow is divided into three major stages:

Stage	Activities
Prepress	Graphic design, typesetting, image assembly, plate making, color separation.
Press	The actual printing operation where ink is transferred to the substrate.
Postpress	Finishing works like cutting, folding, binding, laminating, and packaging.

Module 2: Basic Types of Printing Processes

Duration: 11 Hours | CO2

2.1 Impact vs Non-Impact Printing

- **Impact Printing:** Uses mechanical pressure to transfer ink (e.g., Letterpress, Dot Matrix).
- **Non-Impact Printing:** No physical contact between the print head and the substrate (e.g., Inkjet, Laser/Digital).

2.2 Core Printing Principles

Relief Printing

The image area is **raised** above the non-image area. Examples: Letterpress, Flexography.

Planographic Printing

The image and non-image areas are on the **same plane**. Based on the principle that oil and water don't mix. Example: Offset.

Intaglio Printing

The image area is **recessed** or engraved into the surface. Example: Gravure.

Stencil Printing

Ink is pushed **through** a mesh or stencil. Example: Screen Printing.

Module 3: Workflow and Functional Units

Duration: 13 Hours | CO3

3.1 Offset Printing Units

Offset printing uses three main cylinders:

1. **Plate Cylinder:** Holds the printing plate.
2. **Blanket Cylinder:** A rubber cylinder that picks up the image from the plate.
3. **Impression Cylinder:** Presses the paper against the blanket cylinder.

Dampening System: Applies water to the non-image area.

Inking System: Applies ink to the image area.

3.2 Gravure Printing Units

- **Gravure Cylinder:** Engraved with tiny cells that hold ink.
- **Doctor Blade:** Wipes excess ink from the non-image surface of the cylinder.

- **Impression Roller:** Facilitates ink transfer to the substrate.

3.3 Flexography Units

- **Fountain Roller:** Picks up ink from the reservoir.
- **Anilox Roller:** A ceramic/metal roller that delivers a controlled amount of ink to the plate.
- **Plate Cylinder:** Holds the flexible rubber/photopolymer plate.

Module 4: Screen and Digital Printing

Duration: 9 Hours | CO4

4.1 Screen Printing Equipment

- **Frames:** Wooden, metal, or self-stretching frames.
- **Mesh Materials:** Silk (traditional), Synthetic (nylon/polyester), or Stainless Steel.
- **Squeegee:** A rubber blade used to force ink through the mesh.

4.2 Substrates

Screen printing is versatile and can print on:

- Paper and Paperboard
- Textiles (T-shirts)
- Plastics, Metals, and Wood

- Ceramics and Glass

4.3 Digital Printing Basics

Digital printing eliminates the need for plates. Images are sent directly to the printer from digital files (PDF/TIFF). It is ideal for short runs and variable data printing.

Technical Calculations

Concept: Resolution and DPI

In printing, resolution is measured in Dots Per Inch (DPI).

Formula: Total Pixels = Dimension (inches) × DPI

Example: Calculate the pixels required for a 4x6 inch photo at 300 DPI.
Width: 4 × 300 = 1200 px
Height: 6 × 300 = 1800 px
Total: 1200 × 1800 pixels.

Process Diagrams

Relief Printing: Raised surface holds ink.

Intaglio Printing: Recessed cells hold ink.

Practice Questions

▼ **1. Define Printing. (2 Marks)**

Printing is the industrial process for reproducing text and images using a master template or digital file, typically applying ink to paper or other substrates under pressure or via non-impact methods.

▼ **2. Differentiate between Prepress and Postpress. (5 Marks)**

Prepress: Includes all steps before actual printing, such as design, typesetting, color separation, and plate making.
Postpress: Includes all finishing steps after printing, such as cutting, folding, gathering, binding, and lamination.

▼ **3. Explain the principle of Offset printing. (5 Marks)**

Offset printing is a planographic process based on the chemical principle that oil (ink) and water do not mix. The image and non-image areas are on the same plane. The image is first transferred from the plate to a rubber blanket and then to the paper.

Practice Model Paper

Part A (Short Answer - 2 Marks each)

1. Name two mesh materials used in screen printing.
2. What is the function of a Doctor Blade?
3. Define Impact printing.

Part B (Medium Answer - 5 Marks each)

1. Explain the working of an Anilox roller in Flexography.
2. List the advantages of Digital printing over conventional methods.

Part C (Long Answer - 10 Marks each)

1. Describe the three-cylinder system in an Offset printing machine with a diagram.

Quick Revision

- **Relief:** Letterpress, Flexo (Raised).
- **Planography:** Offset (Flat, Oil/Water).
- **Intaglio:** Gravure (Recessed/Sunken).
- **Stencil:** Screen (Through mesh).
- **Gutenberg:** Father of modern printing (Movable type).
- **Blanket:** Rubber cylinder in offset that makes the process "indirect".

Glossary

Substrate
The base material being printed on (e.g., paper, plastic).

Squeegee
Tool used in screen printing to spread ink.

Anilox
A cylinder used to provide measured ink to a flexo plate.

Official Source Declaration

This content is based on the **SITTTR Kerala Revision 2021** syllabus for the course **Basics of Printing Processes (2101)**. The structure follows the official module distribution and course outcomes provided by the Department of Technical Education, Kerala.

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